

## Creating Your Requirements File

Create a file named `requirements.txt` with the essential libraries:

```
numpy
pandas
matplotlib
scikit-learn
statsmodels
yfinance
backtesting
```

Install these requirements:

```
1. pip install -r requirements.txt
```

## Testing Your Environment

Let's create a simple script to test that everything is working correctly.

Create a file named `test_environment.py` with the following content:

```
1. # test_environment.py
2. import numpy as np
3. import pandas as pd
4. import matplotlib.pyplot as plt
5. import yfinance as yf
6. import datetime
7.
8. # Print library versions
9. print(f"NumPy version: {np.__version__}")
10. print(f"pandas version: {pd.__version__}")
11. print(f"Matplotlib version: {plt.__version__}")
12. print(f"yfinance version: {yf.__version__}")
13.
14. # Download some sample data
15. print("\nDownloading Apple stock data...")
16. start_date = datetime.datetime.now() - datetime.timedelta(days=30)
17. end_date = datetime.datetime.now()
18. apple_data = yf.download('AAPL', start=start_date, end=end_date)
19.
20. # Display the first few rows
21. print("\nApple stock data:")
22. print(apple_data.head())
23.
```